

Business Report

Business Intelligence Report: NYC Taxi Operations Analysis

Course: DSA3050A – Business Intelligence & Data Visualization

Project: Group A – New York City Taxi Rides Analysis

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Team Members

Name	Student ID (Last 3)
Tanveer Omar	762
Mohamed Mohamed	006
Mitchel Muthaura	413
Calvin Gacheru	035
Claire Mwarari	470
Lavender Nchagwa	647

Table of Contents

1. [Executive Summary](#)
2. [Problem Statement](#)
3. [Business Questions](#)
4. [Data Source & Scope](#)
5. [Data Pipeline: ETL & Transformations](#)
6. [Data Architecture: Star Schema Model](#)
7. [DAX Measures & Calculated Logic](#)
8. [Dashboard Overview](#)
9. [Key Findings & Insights](#)
10. [Strategic Recommendations](#)
11. [Limitations & Next Steps](#)
12. [Conclusion](#)
13. [References](#)

1. Executive Summary

New York City's taxi system is one of the world's most complex urban transportation networks. This report presents the findings of a Business Intelligence analysis conducted on over **50,000 NYC taxi trip records**, with the objective of uncovering actionable operational insights for fleet managers, dispatchers, and policy stakeholders.

Key Takeaways:

- **Insight:** Demand is highly concentrated in specific neighborhoods and time windows (peak hours), leaving the fleet over-supplied during off-peak periods.
Impact: Driver idle time and passenger wait times both increase when supply and demand are misaligned.
Action: Implement predictive rebalancing to shift driver availability toward historical demand peaks.
- **Insight:** A significant proportion of trips involve a single passenger, leaving vehicle capacity under-utilized.
Impact: Revenue per mile is suppressed, eroding the fleet's net margin.
Action: Introduce shared-ride incentive tiers to increase per-seat revenue.
- **Insight:** Cash transactions correlate with higher inter-trip idle time compared to digital payments.
Impact: Slower trip turnover reduces total trips-per-hour for drivers accepting cash.
Action: Offer incentives for digital payment adoption to increase operational velocity.

The analysis was conducted using Microsoft Power BI, with data modeled in a star schema and business logic implemented via DAX measures. An interactive dashboard supports real-time, slice-and-filter decision-making across all findings.

2. Problem Statement

NYC's taxi operations face a persistent challenge in **balancing supply and demand**. The mismatch between dynamic passenger demand and active vehicle supply creates two compounding problems:

1. **Driver Idle Time:** Taxis congregate in low-demand areas, causing extended periods without fares and reducing driver earnings.
2. **Passenger Wait Times:** High-traffic neighborhoods experience insufficient taxi coverage, especially during peak hours, leading to passenger abandonment and lost revenue.

To solve this, three critical operational variables must be analyzed and synchronized:

Variable	Description
Location	Geographic hotspots for pickups vs. drop-offs across NYC neighborhoods
Timing	Hourly and day-of-week fluctuations in ride requests
Trip Behavior	Patterns in duration, distance, passenger count, and payment type

Project Goal: To provide data-driven recommendations that improve taxi availability, reduce wait times, and maximize driver revenue through optimized routing and timing decisions.

3. Business Questions

This analysis was structured around seven core business questions that map directly to operational decision-making:

#	Question	Theme
1	What are the peak hours and days for taxi demand across different neighborhoods?	Demand Timing
2	Which neighborhoods generate the highest trip volumes and revenue?	Location Performance
3	How does trip distance correlate with fare amount and payment type?	Fare & Payment
4	What is the average trip duration and how does it vary by time of day?	Operational Efficiency
5	Which payment methods are most common and how do they vary by neighborhood?	Fare & Payment
6	How many passengers typically ride together, and does this affect trip distance?	Vehicle Utilization
7	What is the relationship between pickup and drop-off neighborhoods (commuter patterns)?	Origin-Destination Flow

4. Data Source & Scope

Dataset: New York City Taxi Rides
Source: Kaggle – Surekha Ramireddy (2023)
URL: <https://www.kaggle.com/datasets/surekharamireddy/new-york-city-taxi-rides>

Source Files

File	Type	Description
trips_1.csv	Fact	50,000+ individual trip records
pickup_neighborhoods.csv	Dimension	Neighborhood lookup for trip origins
dropoff_neighborhoods.csv	Dimension	Neighborhood lookup for trip destinations

Key Data Fields

Field	Type	Description
id	String	Unique trip identifier
vendor_id	Integer	Taxi vendor code
pickup_datetime	Datetime	Trip start timestamp
dropoff_datetime	Datetime	Trip end timestamp
passenger_count	Integer	Number of passengers
trip_distance	Float	Distance traveled (miles)
pickup_longitude / latitude	Float	Pickup geolocation
dropoff_longitude / latitude	Float	Drop-off geolocation
payment_type	Integer	Payment code (1=Credit Card, 2=Cash, 3=No Charge, 4=Disputed)
trip_duration	Integer	Duration in seconds
pickup_neighborhood	FK	Neighborhood key from DIM table
dropoff_neighborhood	FK	Neighborhood key from DIM table

5. Data Pipeline: ETL & Transformations

All data preparation was performed using **Power Query (M Language)** inside Microsoft Power BI. The pipeline covers the following transformation categories:

5.1 Initial Setup

- **Promoted Headers:** Set the first row of each CSV as column headers to ensure accurate field identification.
- **Renamed Queries:** Applied consistent naming conventions to align with the star schema design:
 - `trips_1` → `FACT_trips`
 - `pickup_neighborhoods` → `DIM_Pickup_hood`
 - `dropoff_neighborhoods` → `DIM_Dropoff_hood`

5.2 Column Management

- **Removed Redundant Columns:** Dropped `pickup_longitude`, `pickup_latitude`, `dropoff_longitude`, and `dropoff_latitude` from the Fact table, since geographic coordinates exist in the Dimension tables and are redundant in the Fact table.

5.3 Data Type Correction

- Corrected `pickup_datetime` and `dropoff_datetime` from text to DateTime format.
- Changed `passenger_count` from Float to Whole Number.

5.4 Column Splitting & Derivation

- **Date/Time Splitting:** Each datetime column was split into two separate columns:
 - `pickup_date` (Date only) and `pickup_time` (Time only)
 - `dropoff_date` (Date only) and `dropoff_time` (Time only)
- **Time Category (Conditional Column):** A text label column was created to classify each trip by time of day:
 - Morning Peak, Midday, Evening Peak, Night
- **Speed Column (Custom Column):** Derived using the formula:
`speed_mph = trip_distance / (trip_duration / 3600)`
This converts seconds to hours and calculates speed in MPH.

5.5 Data Quality

Issue	Action Taken
Null values in neighborhood ID columns	Filtered out empty rows
Zero-distance trips	Excluded trips where <code>trip_distance = 0</code> (outliers that skew averages)

Issue	Action Taken
Duplicate records	Removed duplicate rows based on the unique <code>id</code> column

5.6 Lookup Enrichment (Merge Queries)

- Pickup Neighborhood Names:** Merged `FACT_trips` with `DIM_Pickup_hood` on `pickup_neighborhood` key to bring in neighborhood name as a readable column.
- Dropoff Neighborhood Names:** Same process applied for drop-off neighborhoods.

5.7 Payment Type Decoding (Replace Values)

Payment codes were decoded into human-readable labels:

Code	Label
1	Credit Card
2	Cash
3	No Charge
4	Disputed

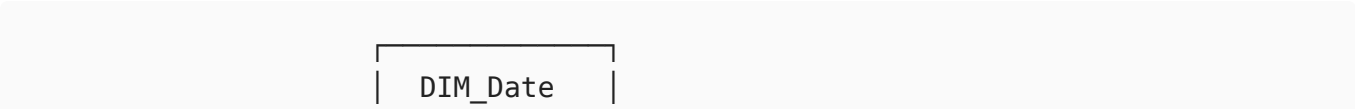
6. Data Architecture: Star Schema Model

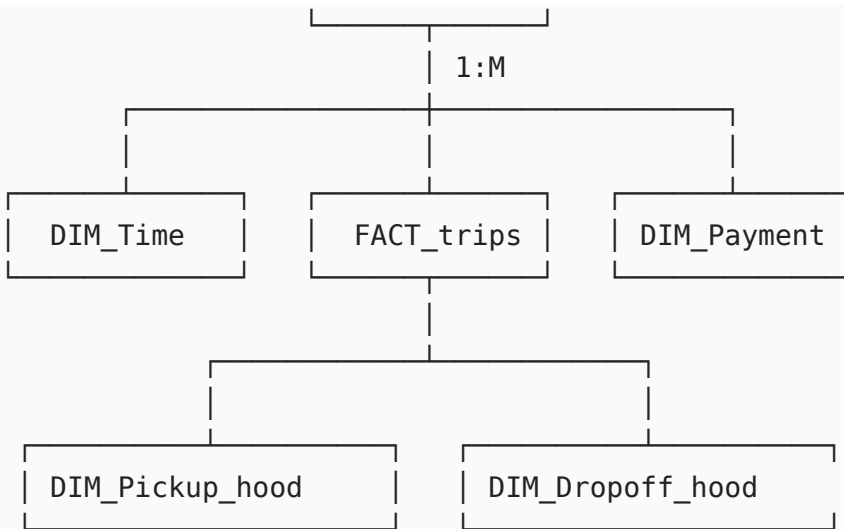
To ensure high query performance and analytical flexibility, the data was structured as a **Star Schema**—a widely adopted dimensional modeling pattern in Business Intelligence.

6.1 Design Rationale

- Fast Analytics:** Denormalized structure minimizes join complexity, enabling Power BI to evaluate measures rapidly.
- Flexible Slicing:** Dimension tables allow filtering by date, time, neighborhood, or payment type without complex query logic.
- Scalability:** New dimensions can be added (e.g., weather, events) without restructuring the Fact table.

6.2 Schema Structure





6.3 Table Definitions

Fact Table: FACT_trips

Column	Description
id	Unique trip key
vendor_id	Taxi vendor
pickup_date / pickup_time	Temporal keys linking to DIM_Date and DIM_Time
passenger_count	Number of passengers
trip_distance	Miles traveled
payment_type	FK to DIM_Payment
trip_duration	Duration in seconds
speed_mph	Derived speed column
pickup_neighborhood	FK to DIM_Pickup_hood
dropoff_neighborhood	FK to DIM_Dropoff_hood

Dimension Tables

Table	Purpose	Key Fields
DIM_Date	Calendar hierarchy	Date, Year, Month, Day, Weekday, Quarter
DIM_Time	Hour-level analysis	Hour, TimeCategory (Peak/Off-Peak/Night)
DIM_Payment	Payment decoding	PaymentCode, PaymentDescription
DIM_Pickup_hood	Pickup geography	neighborhood_id, neighborhood_name

Table	Purpose	Key Fields
DIM_Dropoff_hood	Drop-off geography	neighborhood_id, neighborhood_name

All dimension-to-fact relationships are **one-to-many**, with single-direction cross-filtering.

7. DAX Measures & Calculated Logic

Business logic was implemented using **DAX (Data Analysis Expressions)** in Power BI. The following measures and calculated tables were developed:

7.1 Calculated Dimension Tables

Table	DAX Method	Description
DIM_Date	CALENDAR + ADDCOLUMNS	Auto-generates a full date table with Year, Month, Day, Weekday, Quarter columns
DIM_Payment	DATATABLE	Hardcoded lookup table mapping payment codes (1–4) to descriptive labels

7.2 Core Measures

Measure	DAX Formula	Business Purpose
Total Trips	COUNTROWS(FACT_trips)	Total count of valid taxi rides
Total Distance	SUM(FACT_trips[trip_distance])	Cumulative distance across all trips
Average Passengers	AVERAGE(FACT_trips[passenger_count])	Typical vehicle occupancy per trip
Avg Duration (Mins)	AVERAGE(FACT_trips[trip_duration]) / 60	Mean trip time in minutes

7.3 Time Intelligence Measures

Measure	DAX Logic	Business Purpose
Trips YTD	TOTALYTD([Total_Trips], DIM_Date[Date])	Year-to-date cumulative trip volume

Measure	DAX Logic	Business Purpose
Trips MTD	TOTALMTD([Total_Trips], DIM_Date[Date])	Month-to-date cumulative trip volume
Trips YoY Growth	([Total_Trips] - [PrevYearTrips]) / [PrevYearTrips] using SAMEPERIODLASTYEAR	Year-over-year percentage change in demand
Trips QoQ Change	Current quarter vs. previous quarter using PREVIOUSQUARTER	Quarter-over-quarter demand trend

7.4 Operational & Categorical Measures

Measure	DAX Logic	Business Purpose
Credit Card %	DIVIDE(CALCULATE([Total_Trips], payment_type=1), [Total_Trips], 0)	Share of trips paid by credit card
Cash %	DIVIDE(CALCULATE([Total_Trips], payment_type=2), [Total_Trips], 0)	Share of trips paid in cash
Peak Hour Trips	CALCULATE([Total_Trips], DIM_Time[TimeCategory] = "Peak")	Trip volume during high-demand windows
Speed Category	SWITCH on avg speed: >30 = "Fast", >15 = "Normal", else "Slow"	Traffic congestion classification per trip

8. Dashboard Overview

The Power BI report is structured across three interactive pages, each addressing a distinct analytical lens.

Page 1 – Trip Volume Overview

Purpose: Analyze how different variables drive total trip volume.

KPI Cards:

- Total Distance (cumulative miles)
- Total Trips (count)
- Average Ride Duration (minutes)

Visuals:

- **Pie / Donut Charts:** Proportional breakdown of trips by categorical dimension (e.g., payment type, time category)
 - **Column Chart:** Trip volume distribution across numerical or categorical groups
 - **Table:** Row-level detail with exact metrics per segment
 - **Smart Narrative:** AI-generated text summary of key relationships and trends
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Page 2 – Demand Distribution & Behavior

Purpose: Deep-dive into trip behavior, temporal demand, and geographic patterns.

Interactive Slicers:

- Time Category (Morning, Midday, Evening Peak, Night)
- Payment Method (Credit Card, Cash, No Charge, Disputed)

Visuals:

- **Clustered Column Chart:** Trips by Hour × Time Category — identifies exact peak demand windows
 - **Stacked Column Chart:** Passenger count distribution — reveals single- vs. multi-passenger ride ratios
 - **Donut Chart:** Payment method split — customer preference at a glance
 - **Azure Map:** Pickup location density — geospatial heatmap of high-activity zones
 - **Matrix Table:** Neighborhood performance matrix combining demand volume with trip metrics
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Page 3 – Geographic & Temporal Deep Dive

Purpose: Origin-to-destination flow analysis and hour-by-hour demand mapping.

KPI Cards: Total Distance, Total Trips, Avg Duration (consistent with Page 1)

Interactive Slicers:

- Hours Slicer (0–23, single or multi-select)
- Payment Method Slicer

Visuals:

- **Column Chart:** Total Trips by Hour — pinpoints hourly demand peaks and lulls

- **Pickup Map:** Geospatial view of trip origins across neighborhoods
- **Drop-off Map:** Geospatial view of trip destinations — enables origin-destination flow comparison

Usage Note: Select filters in the slicers and all KPI cards, charts, and maps update dynamically. Hover over map data points or chart columns for detailed tooltips.

9. Key Findings & Insights

Based on the data model, DAX measures, and dashboard analysis, the following findings were identified:

9.1 Demand & Timing

- Taxi demand is strongly concentrated during **morning and evening peak hours**, reflecting commuter patterns. Off-peak hours (late night and midday) see significantly lower volume.
- **Day-of-week effects** are evident, with weekday demand exceeding weekend demand in commuter corridors, while weekend nights show elevated demand in entertainment districts.
- Seasonal variation exists across quarters, as tracked by the **Trips YoY Growth** and **QoQ Change** measures.

9.2 Location & Revenue

- A small number of **high-value neighborhoods** (pickup zones) generate a disproportionate share of total trip volume and distance.
- **Drop-off concentration** in outer neighborhoods creates a "deadhead mileage" problem: drivers returning empty from low-demand drop-off zones lose revenue on those return miles.
- The pickup-to-drop-off neighborhood matrix reveals dominant **commuter corridors** that represent predictable, recurring revenue flows.

9.3 Trip Behavior

- The majority of trips involve **1–2 passengers**, indicating significant unused vehicle capacity (vehicles seat up to 4+ passengers).
- **Trip duration** varies materially by time of day — peak-hour trips take longer for the same distance, reflecting traffic congestion and reducing per-hour driver earnings.
- **Speed Category** analysis shows a meaningful proportion of trips classified as "Slow," corresponding to congested corridors where time-based surcharges could offset profitability loss.

9.4 Payment Patterns

- **Credit Card** is the dominant payment method, with a measurable gap over Cash.
- **Cash transactions** correlate with higher inter-trip idle time, as cash handling adds friction to trip turnover.
- **Disputed and No-Charge** payment types represent a small but notable proportion of records, indicating potential revenue leakage that warrants monitoring.

9.5 Vendor Performance

- Multi-vendor data allows benchmarking: performance differences in efficiency metrics (speed, duration per mile) between vendors suggest that operational protocols differ and that best practices are not uniformly adopted across the fleet.

10. Strategic Recommendations

The following recommendations are prioritized by expected operational impact:

Recommendation 1: Predictive Supply Rebalancing

Finding addressed: Peak demand concentration; driver idle time in low-demand zones.

Action: Deploy algorithmic dispatching that shifts driver positioning toward neighborhoods and hours with historically high demand. Use the DIM_Date and DIM_Time models to generate forward-looking shift schedules aligned with demand curves.

Expected Impact: Reduction in passenger wait times in high-traffic zones; increase in completed trips per driver per shift.

Recommendation 2: Shared-Ride Tier Introduction

Finding addressed: Single-passenger ride dominance; underutilized vehicle capacity.

Action: Introduce an optional shared-ride pricing tier for single-passenger trips in peak-demand corridors. Incentivize passengers with a fare discount while increasing revenue-per-mile for drivers through higher seat utilization.

Expected Impact: Higher revenue per trip mile; improved service capacity without adding vehicles.

Recommendation 3: Digital Payment Incentive Program

Finding addressed: Cash payment friction; lower trips-per-hour for cash-heavy drivers.

Action: Offer minor fare discounts or driver commission bonuses for completed digital payment trips. Reduce the cash-handling overhead that currently increases idle time between trips.

Expected Impact: Increase in overall trips-per-hour metric; faster trip turnover fleet-wide.

Recommendation 4: Congestion Surcharge on Bottleneck Corridors

Finding addressed: Trip duration disproportionately exceeding distance on congested routes.

Action: Identify corridors where the Speed Category is consistently "Slow" and apply dynamic time-based surcharges. Alternatively, establish strategic drop-off points near major transit hubs to reduce driver exposure to congested final miles.

Expected Impact: Revenue protection on slow trips; reduced driver cost on congestion-heavy routes.

Recommendation 5: Vendor Protocol Standardization

Finding addressed: Efficiency gaps between vendors for comparable trip types.

Action: Conduct a structured analysis of the high-performing vendor's dispatch protocols and apply those practices fleet-wide. Use the multi-vendor efficiency benchmarks as the baseline for an operational audit.

Expected Impact: System-wide improvement in average speed and trip duration efficiency, narrowing the performance gap between vendors.

Recommendation 6: Deadhead Mileage Reduction

Finding addressed: Empty return miles from drop-offs in outer, low-demand neighborhoods.

Action: Implement a dynamic return-trip incentive or "backhaul" dispatch system where drivers dropping off in low-demand zones are immediately paired with the nearest available pickup request, even if it requires a small repositioning distance.

Expected Impact: Reduction in total empty miles driven; improvement in net revenue per hour for drivers serving outer zones.

11. Limitations & Next Steps

Current Limitations

Limitation	Detail
Temporal Scope	The dataset covers a defined extract window; trends may not reflect current conditions.
Geolocation Quality	Latitude/longitude precision may vary, affecting neighborhood boundary accuracy.
Revenue Data Absent	Fare amount is not directly present; revenue estimates are inferred from distance and payment type.
External Factors	Weather, public events, and transit disruptions are not included but significantly influence demand.
Vendor Anonymity	Vendor IDs are numeric codes; specific operator identities and their individual practices are unknown.

Recommended Next Steps

1. **Enrich with Weather Data:** Join taxi records with hourly weather data to quantify the demand uplift during rain or extreme temperatures.
2. **Event Calendar Integration:** Overlay major NYC events (concerts, sports, conferences) to model event-driven demand spikes.
3. **Real-Time Pipeline:** Transition from batch CSV analysis to a streaming data pipeline (e.g., Azure Stream Analytics) to power live operational dashboards.
4. **Predictive Modeling:** Apply machine learning (regression or classification) to predict trip demand by neighborhood and hour, enabling proactive fleet positioning.
5. **Fare Amount Integration:** Source or derive actual fare amounts to produce true revenue analytics rather than proxy metrics.

12. Conclusion

This Business Intelligence project successfully transformed over 50,000 raw NYC taxi trip records into structured, actionable insights using a rigorous five-phase approach: data exploration, ETL transformation, dimensional modeling, DAX development, and interactive visualization.

The star schema data model provides a scalable, high-performance analytical foundation. The three-page Power BI dashboard enables stakeholders to explore demand patterns, geographic concentrations, trip behavior, and payment trends through interactive filtering and geospatial mapping.

The central finding is clear: **the NYC taxi fleet's inefficiency is fundamentally a supply-demand alignment problem**, compounded by underutilized capacity and transaction friction. The six strategic recommendations presented in this report offer a practical, data-grounded roadmap for:

- Reducing passenger wait times
- Increasing driver revenue per hour
- Minimizing unprofitable "deadhead" mileage
- Improving fleet-wide operational velocity

By acting on these findings, taxi operators and city planners can move toward a smarter, more equitable, and more profitable urban mobility system.

13. References

Dataset

Ramireddy, S. (2023). *New York City taxi rides* [Data set]. Kaggle.

<https://www.kaggle.com/datasets/surekharamireddy/new-york-city-taxi-rides>

Supplementary Dataset Reference

NYC Taxi Trip Duration Competition Dataset. Kaggle.

<https://www.kaggle.com/datasets/competitions/nyc-taxi-trip-duration/data>

Tool

Microsoft Power BI. <https://powerbi.microsoft.com/>

Interactive Dashboard

[Power BI Dashboard – Group A NYC Taxi Report](#)

Presentation Slides

[Google Slides – NYC Taxi BI Analysis](#)

Report prepared by Group A, DSA3050A – Business Intelligence & Data Visualization.